



$$(x+2)(x^2-2x+4) = x^3+8$$

$$(a+b)^3 = (a+b)(a+b)(a+b)$$

$$\begin{aligned} & a^2+ab+ab+b^2 = a^2+2ab+b^2 \\ & (a^2+2ab+b^2)(a+b) = a^3+a^2b+2a^2b+2ab^2+b^2a+b^3 \\ & = a^3+3a^2b+3b^2a+b^3 \end{aligned}$$

$$(a-b)^3 = (a-b)(a-b)(a-b)$$

$$a^3-3a^2b+3b^2a-b^3$$

$$\begin{aligned} & (a^2-2ab+b^2)(a-b) = \\ & a^3-ba^2-2a^2b+2ab^2+b^2a-b^3 \end{aligned}$$

$$(x-2y)^3 = x^3-2xy-2x^2y+2xy^2+8y^3$$

$$(x-2y)(x-2y) = x^2-4xy+4y^2$$

$$\begin{array}{l} a = x \\ b = 2y \end{array} \quad \left| \quad \begin{array}{l} x^3 - 3 \cdot x^2 \cdot 2y + 3 \cdot (2y)^2 \cdot x - (2y)^3 \\ = x^3 - 6x^2y + 12y^2x - 8y^3 \end{array} \right.$$

$$(x^2 - 4xy + 4y^2)(x - 2y) =$$

$$x^3 - 2x^2y - 4x^2y + 8xy^2 + 4y^2x - 8y^3$$

$$(x+2)(3x-2) - (x-5)(x+2) \quad \text{нредр.} \quad \textcircled{=}$$

$$\sqrt{3x-8} = 5$$

$$3x-8 \geq 0 \rightarrow \begin{array}{l} 3x \geq 8 \\ x \geq \frac{8}{3} \end{array}$$

Область
допуска
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$$3x-8 = 25$$

$$3x = 33$$

$$\underline{X = 11} \quad \textcircled{1A}$$

$$(\sqrt{a})^2 = a$$

$$\left(\sqrt{10 - x} \right)^2 = (3)^2$$

$$10 - x = 9$$

$$x = 1$$

$$\sqrt{13 + 2x} = 5$$

$$\frac{1}{\sqrt{x}} = \frac{1}{5}$$

$$\sqrt{x} = 5$$

$$x = 25$$

Токасетелне ур-а

$$a^x = a^y$$

$$x = y$$

$$2^{2x-3} = 2^{x-2}$$

①

$$2x-3 = x-2$$

$$x = 1$$

$$3x = 3$$

$$2^{\underline{4-2x}} = \underline{64} = 2^6$$

$$4-2x = 6$$

$$\cancel{-2x} = \left(\frac{2}{-2} \right) = -1$$

$$x = -1$$

$$9^{-5+x} = 729$$

$$x = 8$$

$$\frac{3^{5x}}{3^{3x}}$$

$$(5^{x-7}) = \frac{1}{125} = \frac{1}{5^3} = 5^{-3}$$

$$x-7 = -3$$

$$x = 4$$

$$\left(\frac{1}{8}\right)^{-3+x} = 512$$

$$(8^{-3})^{(-3+x)} = 8^3$$

$$8^{3-x} = 8^3$$

$$3-x = 3$$

$$x = 0$$

$$\left(\frac{1}{9}\right)^{2+x} = 729$$

$$6^{2x-6} \cdot 6^{5-3x} = 216 = 6^3$$

$$6^{2x-6+5-3x} = 6^3$$

$$6^{-x-1} = 6^3$$

$$-x-1 = 3$$

$$x = -4$$

$$2^{-3x+1} \cdot 2^{-x-5} = \frac{1}{64} = 2^{-6}$$

$$2^{\underline{-3x+1-x-5}} = 2^{-6}$$

$$2^{-4x-4} = 2^{-6}$$

$$-4x-4 = -6$$

$$-4x = -2$$

$$x = \frac{2}{4} = \frac{1}{2}$$

$$3^{2x-5} \cdot 3^{2x-3} = \frac{1}{81} = 3^{-4}$$

$$3^{2x-5+2x-3} = 3^{-4}$$

$$3^{4x-8} = 3^{-4}$$

$$4x-8 = -4$$

$$\underline{x=1}$$

$$3^x \div 3^{2x+1} = 3^{x-(2x+1)}$$